

AY: 2025-2026
Course: PPOM 2
Week 11, Lecture 103

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DPR Lecture: Abnormal Respiratory Exam

Presented by:

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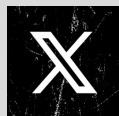
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Session Objectives

- Appreciate the difference between normal and abnormal breath sounds
- Review Physical Exam techniques, along with the “How” and “Why” we do these techniques
- Review where normal breath sounds are heard.
- Recognize the abnormal findings of the thorax and lung exam
- A few brief clinical scenarios with lung sounds to strengthen the basic understanding of the abnormal lung sounds
- Apply critical thinking into forming a differential diagnosis based on history and physical examination.

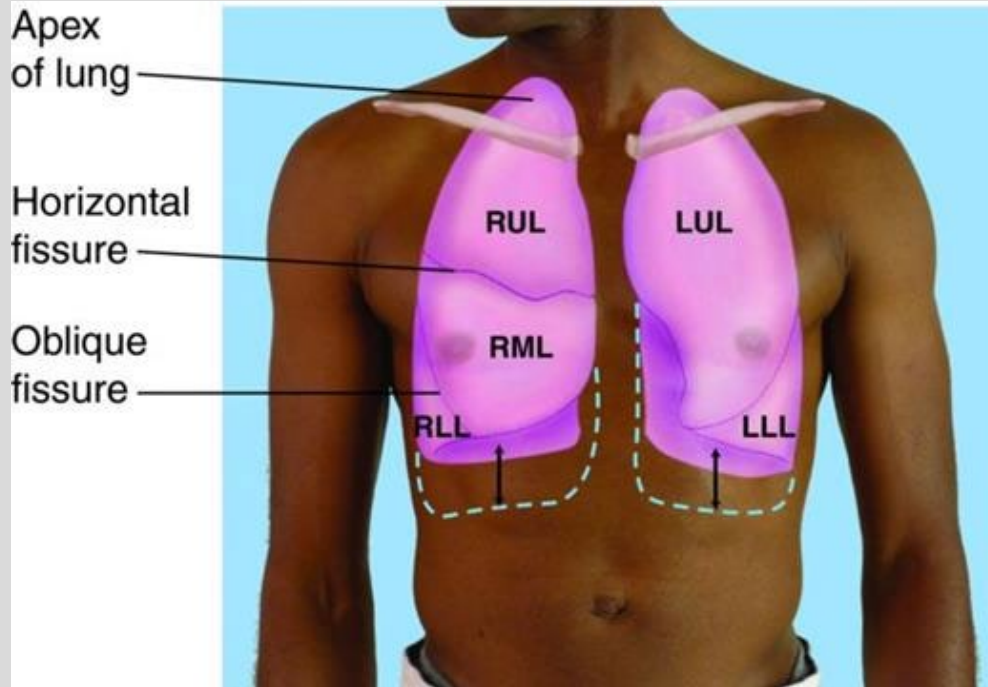
RED WOLVES and BEARS

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Anatomy – Lung Parenchyma

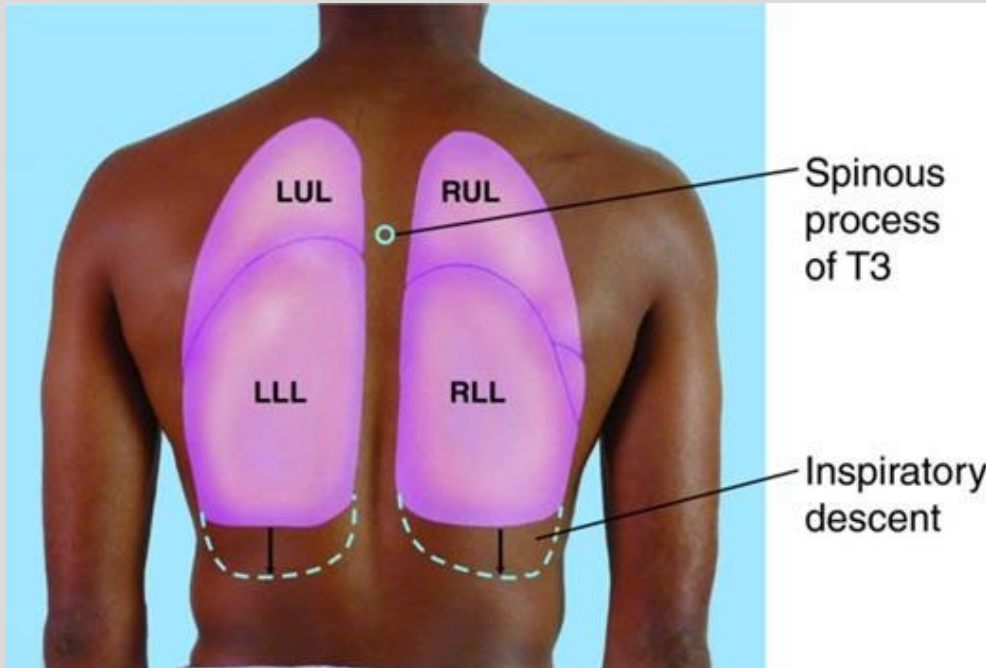


- Anterior View:
 - Right Lung (3 Lobes):
 - Upper, Middle, and a small section of lower lobe
 - Left Lung (2 Lobes):
 - Upper, and a small section of lower lobe

Anatomy – Lung Parenchyma

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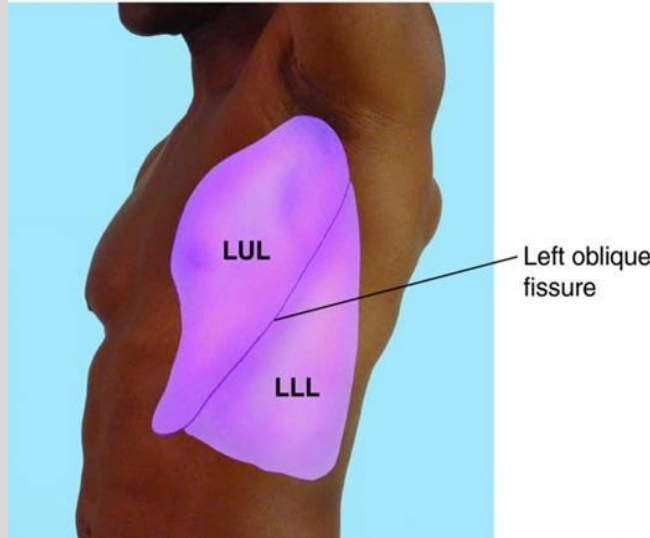
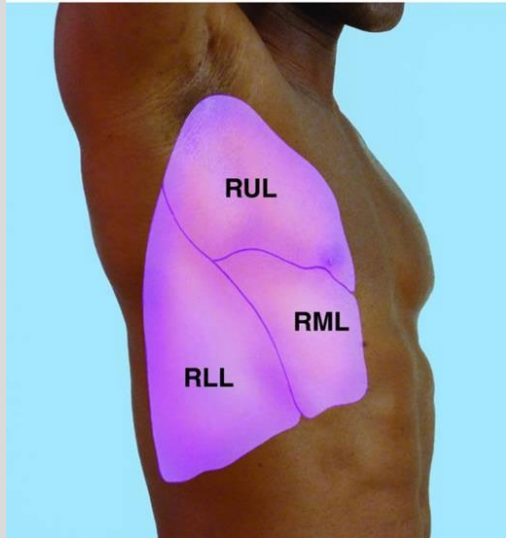
- **Posterior View:**
 - **Right Lung (3 Lobes):**
 - Upper, Lower, and a small section of middle lobe
 - **Left Lung (2 Lobes):**
 - Upper and lower lobes

Anatomy – Lung Parenchyma

Lateral Views

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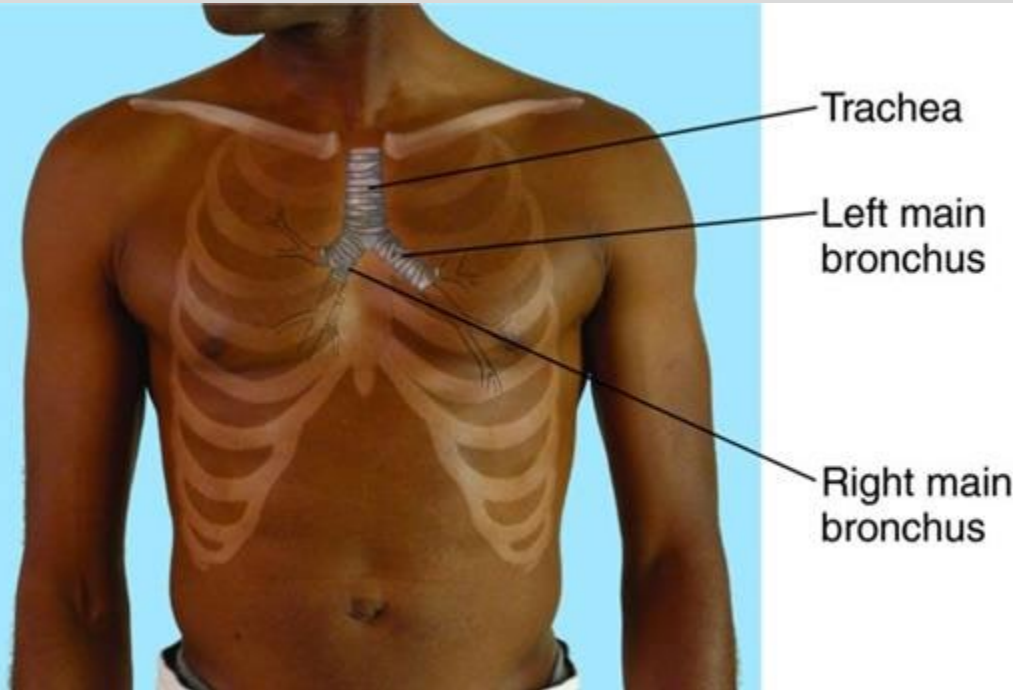


Bickley, Lynn S.; Szilagyi, Peter G., "Bates' Guide to Physical Examination and History Taking", 12e, Wolters Kluwer Health. 2017

Anatomy – Trachea and Major Bronchi

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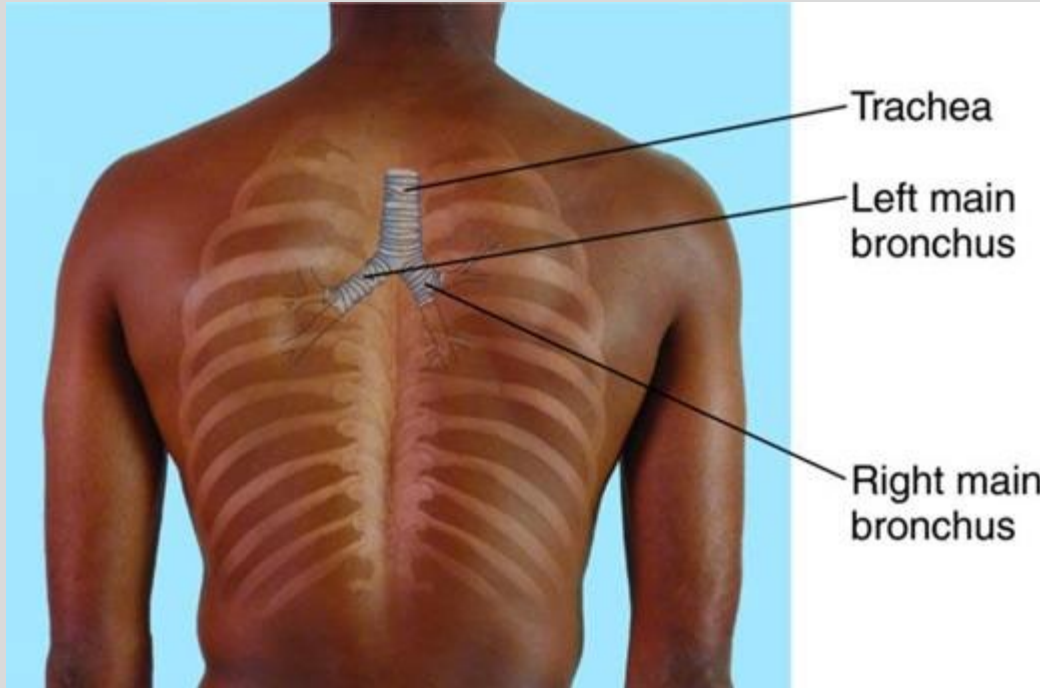


- Anterior View
 - Trachea and Main Stem Bronchi

Anatomy – Trachea and Major Bronchi

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- Posterior View
 - Trachea and Main Stem Bronchi

The Health History

- Common or Concerning Symptoms
 - Chest Pain
 - Shortness of breath
 - Wheezing
 - Cough
 - Blood-streaked sputum (Hemoptysis)
 - Daytime sleepiness or snoring and disordered sleep

Chest Pain

- Investigate thoracic and cardiac causes
- Is the patient having chest pain with shortness of breath?
- Is the having shortness of breath with chest pain?
- Cardiac causes
 - Angina Pectoris
 - MI
 - Pericarditis
 - Aortic Dissection

Chest Pain

- Pulmonary causes
 - Pleuritic Pain
- GI Causes
 - Gastrointestinal Reflux Disease
 - Diffuse Esophageal Spasm
- Musculoskeletal
 - Chest Wall Pain, Costochondritis
- Psychiatric
 - Anxiety
 - Panic Disorder

Shortness of Breath (Dyspnea) and Wheezing

- Painless but uncomfortable awareness of breathing
- “Have you had any difficulty breathing?”
- Does the symptoms occur at rest or with exertion
- How much exertion produces onset
- There is no absolute scale for quantifying shortness of breath
- Determine its severity based on the patient’s daily activities

Cough

- A reflex response to stimuli that irritate receptors in the larynx, trachea, or large bronchi
- Stimuli include mucus, pus, and blood
- Stimuli can be external agents
- Other causes
 - Inflammation of the respiratory mucosa
 - Pneumonia
 - Pulmonary edema
 - Tumor or enlarge peribronchial lymph nodes
 - Cardiac causes (CHF)

Cough – Establish Duration

- Acute cough (less than 3 weeks)
 - Most common cause is Viral Upper Respiratory Infection
 - Acute bronchitis
 - Pneumonia
 - Left-Sided heart failure
 - Asthma
 - Foreign body
 - Smoking
 - ACE-I therapy
- Subacute (3 to 8 weeks)
 - Post-Infectious
 - Pertussis
 - Acid reflux
 - Asthma
 - Bacterial Sinusitis
- Chronic (more than 8 weeks)
 - Postnasal Drip
 - Asthma
 - GERD
 - Chronic Bronchitis
 - Bronchiectasis

Cough

- Ask whether the cough is dry or produces sputum, or phlegm
 - Mucoid sputum → Viral infections and cystic fibrosis
 - Purulent sputum → often accompanies bacterial pneumonia
- Describe the volume of any sputum and its color, odor, and consistency
- “How much do you think you cough up in 24 hours: a teaspoon, tablespoon, quarter cup, half cup, cupful?”

Hemoptysis

- Blood coughed up from the lower respiratory tract
- Quantify the volume of blood produced, the setting and activity, and any associated symptoms
- Try to confirm the source of the bleeding

Daytime Sleepiness or Snoring and Disordered Sleep

- May report excessive daytime sleepiness and fatigue
- Ask about problems with snoring, witnessed apneas, awakening with a choking sensation, or morning headache.
- Consider Obstructive sleep apnea, which is commonly seen in:
 - Obesity
 - Treatment-resistant hypertension
 - Posterior malocclusion of the jaw (retrognathia)
 - Heart Failure
 - Atrial Fibrillation
 - Stroke
 - Type 2 diabetes

Daytime Sleepiness or Snoring and Disordered Sleep

- Mechanisms include:
 - Instability of the brainstem respiratory center
 - Disordered sleep arousal
 - Disordered contraction of upper airway muscles (genioglossus malfunction)
 - Anatomic changes contributing to airway collapses such as obesity

Physical Exam

Physical Exam

- Respiratory Rate (and all other vitals)
- Inspection
- Palpation
- Percussion
- Auscultation
- Additional examination skills
 - Fremitus, egophony, whispered pectoriloquy, bronchophony

Rate and Rhythm of Breathing

- Respiratory Rate is about 14-20 per minute in adults, up to 44 per minute in infants
- Slow Breathing (Bradypnea)
 - Abnormal alveolar hypoventilation without increased tidal volume can arise from
 - Uremia
 - Drug-Induced respiratory depression
 - Increased Intracranial pressure
- Sighing Respiration
 - Breathing punctuated by frequent sighs suggests Hyperventilation Syndrome
 - A common cause of dyspnea and dizziness
 - Occasional sighs are normal

Rate and Rhythm of Breathing

- Rapid Shallow Breathing (tachypnea)
 - Salicylate intoxication
 - Restrictive lung disease
 - Pleuritic chest pain
 - Elevated diaphragm
- Cheyne-Stokes Breathing
 - Periods of deep breathing alternate with periods of apnea (no breathing).
- Obstructive Breathing
 - Expiration is prolonged due to narrowed airways increase the resistance to air flow
 - Asthma, COPD, Chronic Bronchitis

Rate and Rhythm of Breathing

- Rapid Deep Breathing
 - Hyperpnea
 - Rapid deep breathing occurs in response to metabolic demands
 - Exercise
 - High Altitude
 - Sepsis
 - Anemia
 - Hyperventilation
 - Independent of metabolic demand, except in respiratory acidosis
 - Light-headedness and tingling may arise from decreased CO₂ concentration

Rate and Rhythm of Breathing

- Kussmaul breathing
 - Compensatory over breathing due to systemic acidosis
- Ataxic Breathing (Biot Breathing)
 - Irregular breathing
 - Periods of apnea alternate with regular deep breaths which stop suddenly for short intervals
 - Causes include meningitis, respiratory depression, and brain injury, typically at the medullary level.

Signs of Respiratory Distress

- Assess the respiratory rate
- Inspect the patient's color for cyanosis or pallor
- Listen for audible sounds of breathing
- Inspect the neck for contraction of accessory muscles during inspiration, or contraction of the intercostal or abdominal oblique muscles during expiration
- Is the Trachea midline

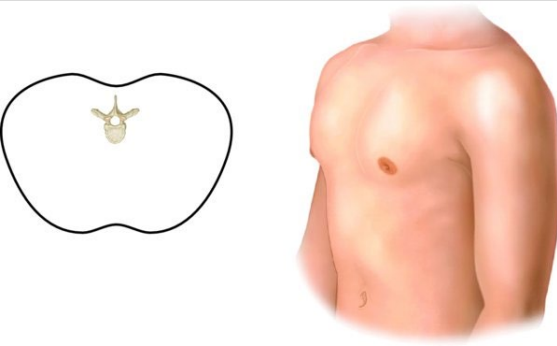
Examination of the Chest

- **Inspection-**

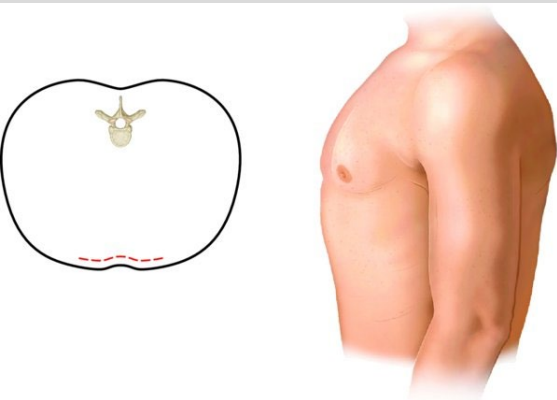
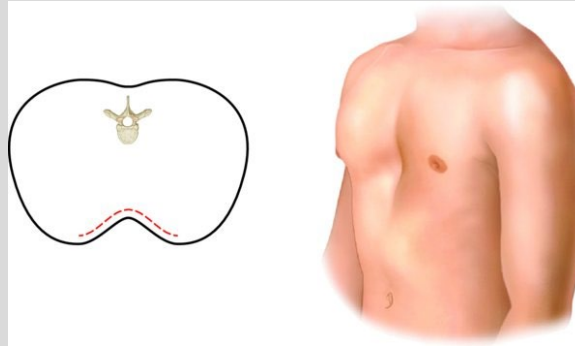
- Note the shape of the chest and how the chest moves
 - Funnel Chest (Pectus Excavatum)
 - Barrel Chest
 - Pigeon Chest (Pectus Carinatum)
 - Traumatic Flail Chest
 - Thoracic Kyphoscoliosis

Deformities of the Thorax

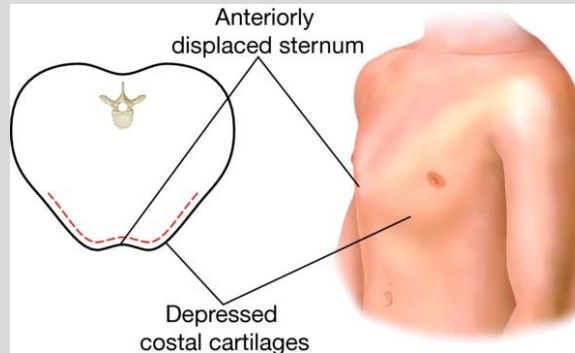
Normal Adult



Funnel Chest (Pectus Excavatum)



Barrel Chest



Pigeon Chest (Pectus Carinatum)

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Bickley, Lynn S.;
Szilagyi, Peter G.,
“Bates' Guide to
Physical Examination
and History Taking”,
12e, Wolters Kluwer
Health. 2017

Pectus Carinatum

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The
sternum is
displaced
anteriorly,
increasing
the AP
diameter



<https://giphy.com/gifs/pigeon-taking-GJp5MvZlsQTbW>

<http://www.orthokids.com.au/conditions-treated/scoliosis/pectus-carinatum>

Pectus Excavatum



<https://olympics.com/en/news/cody-miller-iron-man-olympic-swimming>

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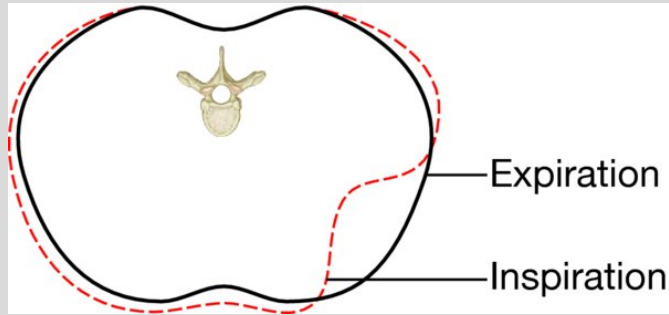
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- Funnel Chest or “Sunken Chest
- - The sternum is caved-in appearance
- Can impact cardiac output
- Can decrease lung capacity

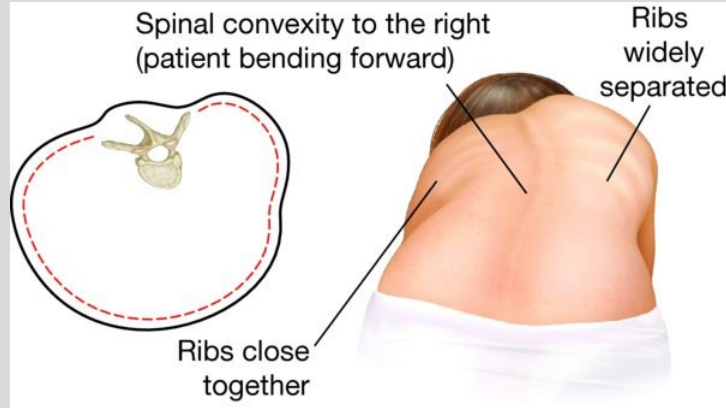
Deformities of the Thorax

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Traumatic Flail Chest



Thoracic Kyphoscoliosis

Bickley, Lynn S.; Szilagyi, Peter G., "Bates' Guide to Physical Examination and History Taking", 12e, Wolters Kluwer Health. 2017

Examination of the Chest

- **Inspection-**

- Note the shape of the chest and how the chest moves
 - Deformities or asymmetry in chest expansion
 - Abnormal muscle retraction of the intercostal spaces during inspiration
 - Impaired respiratory movement on one or both sides or a unilateral lag in movement

Examination of the Chest

- **Palpation-**
 - Focus on areas of tenderness or bruising, respiratory expansion, and fremitus
 - Identify tender areas
 - Note any palpable crepitus or “step-offs”
 - Evaluate for chest expansion
 - Palpate both lungs for symmetric tactile fremitus

Examination of the Chest

- Palpation

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Examination of the Chest- Palpation Tactile Fremitus

- Abnormal findings:
 - Decreased Fremitus:
 - Conditions where air is “trapped” in lung space.
 - *COPD, asthma, atelectasis*
 - Conditions which increase the space between lung and chest wall.
 - *Pleural effusion, pneumothorax*
 - Increased Fremitus:
 - Conditions where solid/fluid are found in the lung.
 - *Consolidation, some tumors*

Examination of the Chest

- **Percussion-**

- Produces audible sound and palpable vibrations
- Helps you establish whether the underlying tissues are air-filled, fluid-filled, or consolidated
- Healthy lungs are resonant
- Dullness → Fluid or solid tissue replaces air-containing lung or occupies the pleural space beneath your percussing finger
 - Examples include lobar pneumonia and pleural effusion

Examination of the Chest

- **Percussion-**

- Hyperresonance

- Generalized → Common over the hyperinflated lungs of COPD or Asthma
- Unilateral hyperresonance → Suggests a large pneumothorax or an air-filled bulla

Examination of the Chest

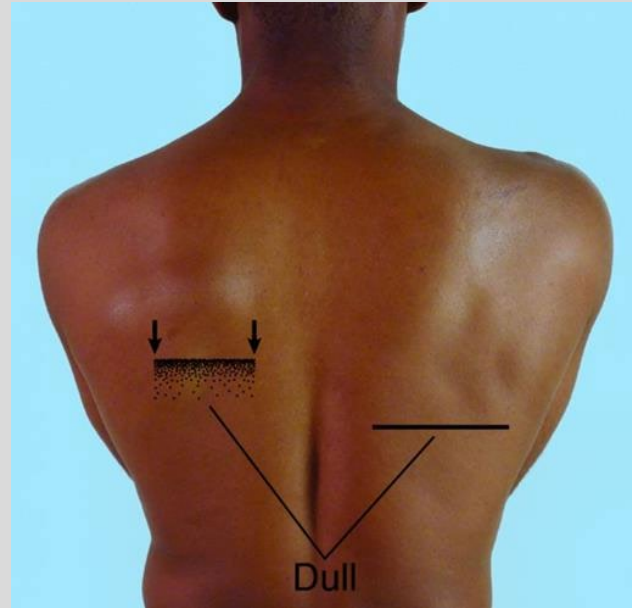
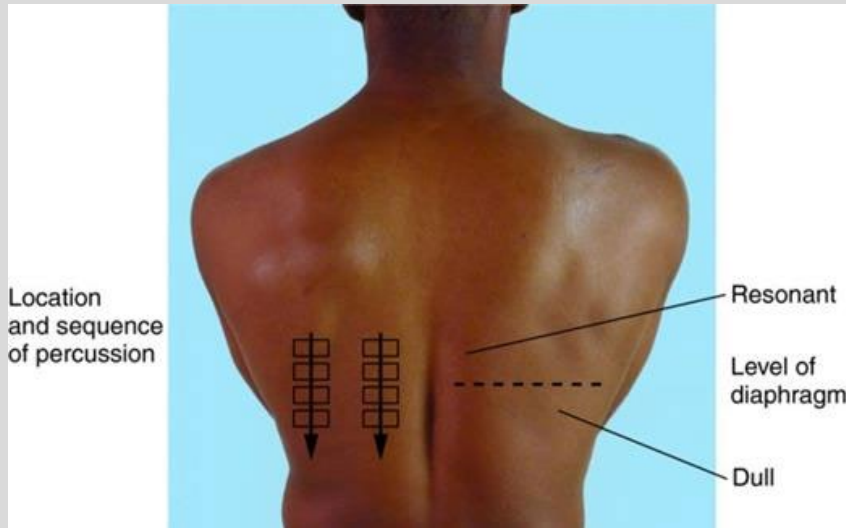
- **Percussion-**

- Dullness

- Allows to identify the descent of the diaphragm, or diaphragmatic excursion
- Begin percussion above the expected level of dullness
- Percuss downward in progressive steps until dullness clearly replaces resonance
- Confirm level of change by percussing downward from adjacent areas both medially and laterally

Examination of the Chest

- Percussion



Bickley, Lynn S.; Szilagyi, Peter G., "Bates' Guide to Physical Examination and History Taking", 12e, Wolters Kluwer Health. 2017

An abnormally high level suggests a pleural effusion or an elevated hemidiaphragm from atelectasis or phrenic nerve paralysis

Examination of the Chest

- **Percussion-**

- Dullness

- Identifying the boundary between the resonant lung tissue and the duller structures below the diaphragm
- Estimate the extent of diaphragmatic excursion
 - Determine the distance between the level of dullness on full expiration and the level of dullness on full inspiration
 - Normal is about 3 to 5.5 cm

Examination of the Chest

- **Auscultation-**

- Assess air flow through the tracheobronchial tree
- Listening to the sounds generated by breathing
- Listening for any adventitious (added) sounds
- If abnormalities are suspected, listening to the sounds of the patient's spoken or whispered voice as they are transmitted through the chest wall

Examination of the Chest

- **Auscultation-**

- Breath sounds may be decreased
 - When air flow is decreased
 - Obstructive lung disease or respiratory muscle weakness
 - When the transmission of sound is poor
 - Pleural effusion, pneumothorax, COPD

Examination of the Chest

- **Auscultation-**

- Listen for Normal Breath Sounds (Lung Sounds)
 - Vesicular → Soft and low-pitched. Throughout inspiration, continue without pause through expiration, and fade away about 1/3 of the way through expiration
 - Bronchovesicular → Inspiratory and expiratory sounds about equal in length, at times separated by a silent interval
 - Bronchial → Louder, harsher and higher in pitch, with a short silence between inspiratory and expiratory sounds
 - Tracheal → Loud harsh sounds heard over the trachea in the neck
- If Normal Breath Sounds heard in Abnormal Place, then suspect pathology
 - Ex. Bronchovesicular or bronchial breath sounds in abnormal place suggest replacement of air-filled lungs by fluid-filled or consolidated lung tissue.

Examination of the Chest

- **Auscultation-**

- Tracheal Breath Sound

- Very loud, high-pitched
- Heard over the trachea
- Inspiratory and Expiratory sounds are almost equal



- Vesicular Breath Sound

- Soft, low-pitched
- Hear over most of the lungs
- Longer in Inspiration vs. Expiration



Examination of the Chest

- **Auscultation-**

- Bronchial Breath Sound

- Loud, high-pitched
- Heard over the manubrium (Larger proximal airways)
- Longer in expiration vs. inspiration



- Bronchovesicular Breath Sound

- Intermediate in intensity and pitch
- Heard over the 1st and 2nd interspaces anteriorly and between the scapulae posteriorly
- Heard equally in inspiration and expiration



Normal Findings

• Characteristics of Breath Sounds¹⁶

	Duration of Sounds	Intensity of Expiratory Sound	Pitch of Expiratory Sound	Locations Where Heard Normally
Vesicular* 	Inspiratory sounds last longer than expiratory ones.	Soft	Relatively low	Over most of both lungs
Broncho-vesicular 	Inspiratory and expiratory sounds are about equal.	Intermediate	Intermediate	Often in the 1st and 2nd interspaces anteriorly and between the scapulae
Bronchial 	Expiratory sounds last longer than inspiratory ones.	Loud	Relatively high	Over the manubrium, if heard at all
Tracheal 	Inspiratory and expiratory sounds are about equal.	Very loud	Relatively high	Over the trachea in the neck

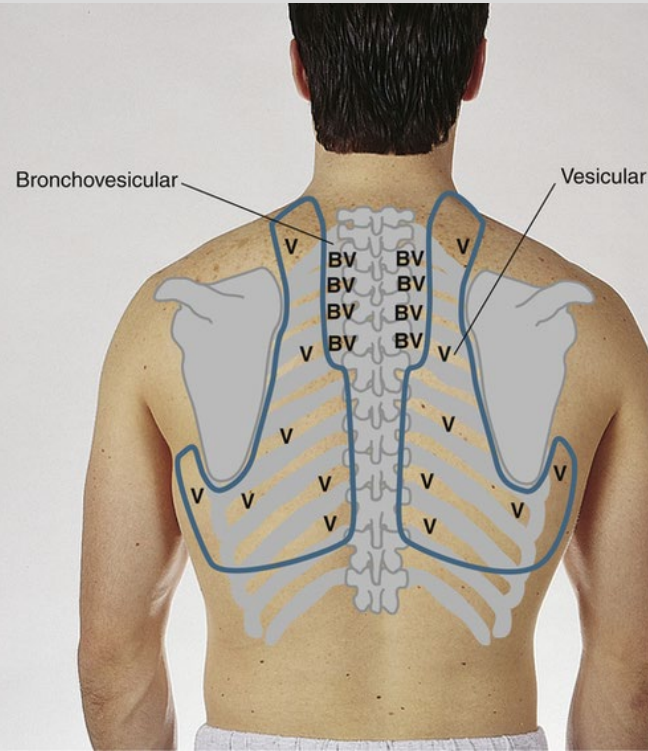
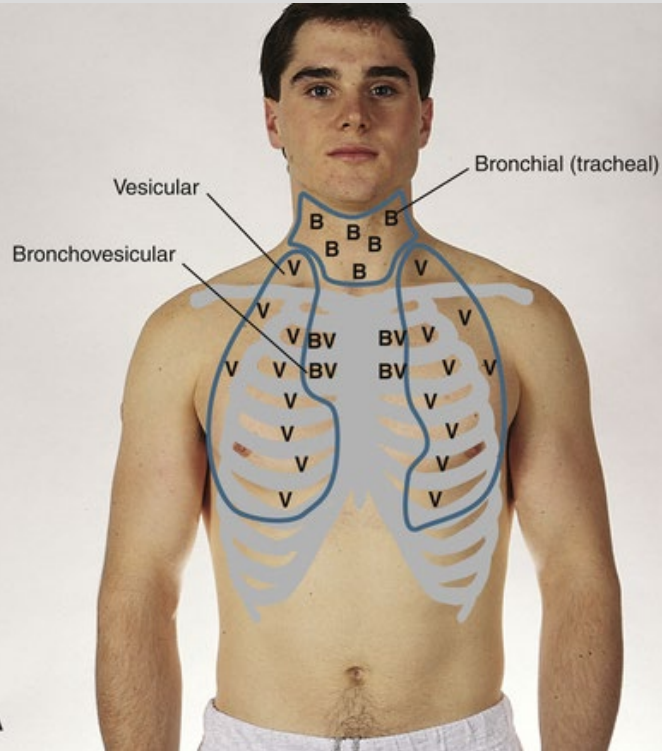
* The thickness of the bars indicates intensity; the steeper their incline, the higher the pitch.

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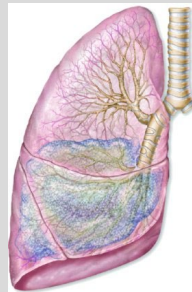
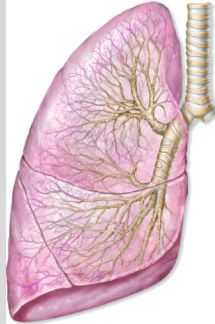
Normal Findings

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Modified from
Jarvis C: Physical
examination &
health assessment,
ed 5, Philadelphia,
2008, Saunders.



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Breath Sounds

Predominantly vesicular

Bronchial or bronchovesicular
over the involved area

Transmitted Voice Sounds

Spoken words muffled and
indistinct
Spoken "ee" heard as "ee"
Whispered words faint and
indistinct, if heard at all

Spoken "ee" heard as "ay"
(*egophony*)
Spoken words louder
(*bronchophony*)
Whispered words louder, clearer
(*whispered pectoriloquy*)

Tactile Fremitus

Normal

Increased

NOTE: In the hyperinflated lung
of COPD, breath sounds are
decreased (muffled to distant) to
absent and transmitted voice
sounds and fremitus are
decreased.

NOTE: In the dull lung of pleural
effusion, breath sounds are
decreased to absent (bronchial
sounds possible at upper margin
of effusion). Transmitted voice
sounds are decreased to absent
(but may be increased at upper
margin of effusion). Fremitus is
decreased.

Bickley, Lynn S.;
Szilagyi, Peter G.,
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2017

Examination of the Chest

- **Auscultation-**

- Abnormal breath Sounds
 - Absent or Decreased Breath Sounds
 - COPD/emphysema, pleural effusion, pneumothorax, asthma
 - Bronchial Breath Sounds in Abnormal Locations
 - Indicates area of consolidation
 - Further testing may confirm your suspicions

Examination of the Chest

- **Auscultation- Adventitious Breath Sounds**

- Crackles
 - Results from a series of tiny explosions when small distal airways, deflated during expiration, pop open during inspiration
 - Recent acoustic studies indicate that the role of secretions as a cause of crackles is less likely
 - Discontinuous
 - Intermittent, nonmusical, and brief
 - Heard most commonly on inspiration

Examination of the Chest

- **Auscultation- Adventitious Breath Sounds**

- Crackles
 - Fine Crackles
 - Softer, higher pitched, more frequent
 - Mid to late inspiration
 - Shorter duration and higher frequency
 - “Sudden inspiratory opening of small airways held closed by surface forces during the previous expiration”
 - Coarse Crackles
 - Early inspiration and last throughout expiration (biphasic)
 - Have a popping sound
 - Heard over any lung region
 - Longer duration and lower frequency
 - “boluses of gas passing through airways as they open and close intermittently”

Examination of the Chest

- **Auscultation- Adventitious Breath Sounds**

- Crackles



- Fine Crackles

- Roll a strand of hair between your fingers close to you ear

- Coarse Crackles



- Sounds like a Velcro fastener

Examination of the Chest

- **Auscultation- Adventitious Breath Sounds**

- Wheezes
 - Continuous musical sounds that occur during rapid airflow when bronchial airways are narrowed almost to the point of closure
 - Inspiratory, expiratory, or biphasic (more common during expiration)
 - May be localized or heard throughout the lung
 - Typical of asthma



Examination of the Chest

- **Auscultation- Adventitious Breath Sounds**

- Rhonchi
 - Suggest secretions in the larger airways
 - Considered by some to be a variant of wheezes but lower in pitch
 - May disappear with coughing (so secretions may be involved)
 - Continuous, low-pitched, snoring quality



Examination of the Chest

- **Auscultation- Adventitious Breath Sounds**

- Stridor
 - Caused by partial obstruction of the upper airway
 - Loud, High-pitched
 - Heard during inspiration
 - Can be heard without a stethoscope
 - A sign of respiratory distress



Examination of the Chest

- **Auscultation- Adventitious Breath Sounds**

- Pleural Rub

- Caused by inflamed viscera pleural surfaces as it slides against the parietal pleura
- Discontinuous, low-frequency, grating sound
- Nonmusical, biphasic
- Heard during inspiration and expiration
- Best heard in the axilla and base of the lungs



Auscultation Transmission of Voice Sounds

- Some, if not all, of the following tests can be performed if there is an index of suspicion for consolidation
 - Bronchophony
 - Egophony
 - Whispered Pectoriloquy
- You do not need to perform these tests if not suspecting consolidation

Evaluate Bronchophony

- Patient in seated position.
- Place stethoscope (diaphragm) over posterior lung fields.
- While auscultating, ask patient to say (not whispered) “1-2-3, 1-2-3.” Or “99, 99.”
- Evaluate side to side, at least 2 of 4 levels.
- Sound should be muffled indistinct
- Abnormal is louder voice sounds

Evaluate Egophony

- Patient in seated position.
- Place stethoscope (diaphragm) over posterior lung fields.
- Auscultate on skin
- Ask patient to say “eee.”
- Evaluate side to side, at least 2 of 4 levels.
- Normal is muffled “eee” sound
- Abnormal is if “eee” sounds like “A”

Evaluate Whispered Pectoriloquy

- Patient in seated position.
- Place stethoscope (diaphragm) over posterior lung fields.
- Auscultate on skin
- While auscultating, ask patient to whisper “1-2-3, 1- 2-3.” Or “99, 99.”
- Evaluate side to side, at least 2 of 4 levels.
- Should sound be faint if heard at all
- Abnormal is louder, clearer whispered sounds

Auscultation Transmission of Voice Sounds

All of the following tests become abnormal when there is a consolidation present:

- **Bronchophony:**
 - Positive Findings: as patient says “99”, sounds are NOT muffled, but loud and clear.
- **Egophony:**
 - Positive Findings: as patient says “E”, sounds like the long vowel “A”.
- **Whispered Pectoriloquy:**
 - Positive Findings: as patient whispers “99”, sounds are loud and clear.v

Egophony

- Normal

- “Eee” sounds like “Eee”



- Abnormal

- “Eee” sounds like “Aaa”



Whispered Pectoriloquy

- Normal
 - Whispered numbers (“1,2,3”) sound soft
- Abnormal
 - Whispered numbers (“1,2,3”) are heard louder and clearer



What DO you hear?





What DO you hear?



Clinical Case Studies

Case #1

- 75 y/o male with cough
- Started 6 months ago, worsening this past week. Worse with exertion
- Some white sputum, but most of the time dry
- Associated SOB, no fever
- + smoking history x 50 years (1.5 packs per day), quit 6 months ago



Case #2

- 62 y/o female with cough
- Started 2 weeks ago, dry in nature, worse at night
- Nasal discharge and congestion
- Took OTC allergies medications with improvements
- No fever, shortness of breath
- No smoking history



Case #3

- 18 y/o female with cough
- Started about 24 hours, associated with fever, phlegm
- Worsening and associated with shortness of breath
- OTC meds with no relief
- Physical exam reveals the following lung sound over the Right Middle Lobe

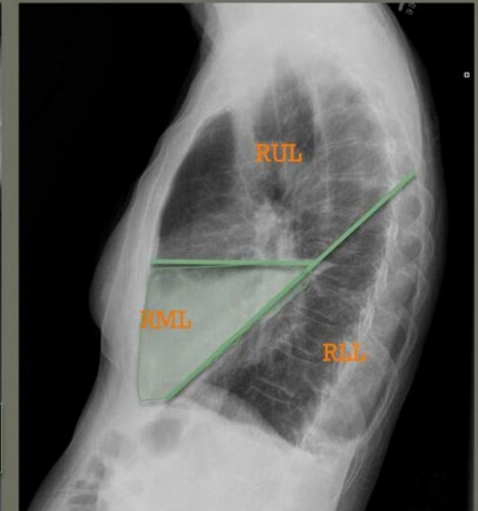
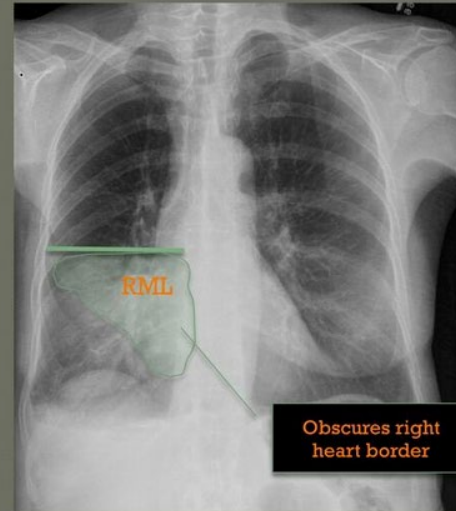


Case #3

- 18 y/o female with cough
- Started about 24 hours, associated with fever, phlegm
- Worsening and associated with shortness of breath
- OTC meds with no relief



Pneumonia II RML



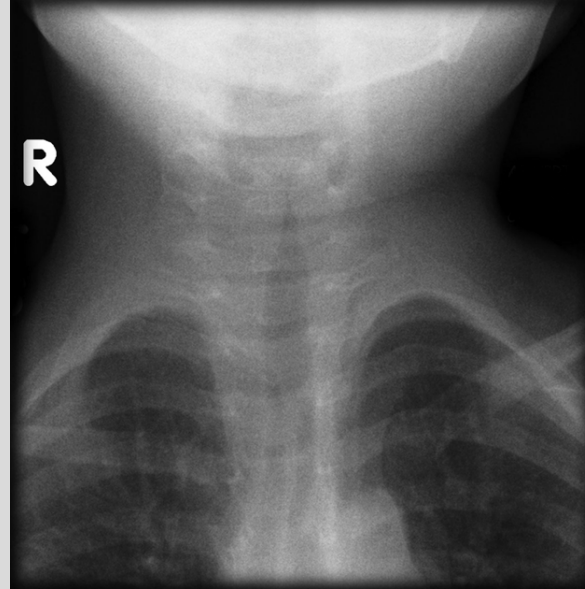
Case #4

- 2 y/o male with a cough
- Started about 5 days ago as a upper respiratory infection, and now worsening over past 24 hours
- Mom reports “he sounds like a seal”
- Very uncomfortable at night



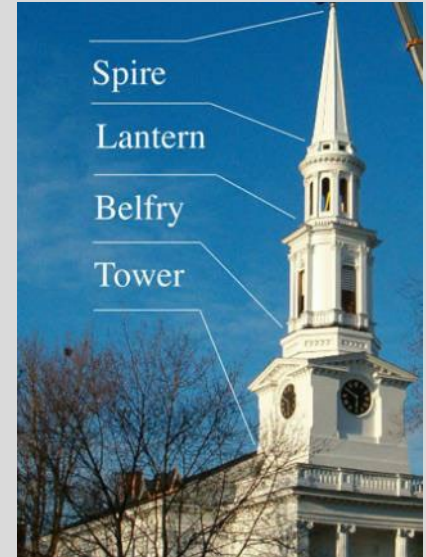
Case #4

- 2 y/o male with a cough
- Started about 5 days ago as a upper respiratory infection, and now worsening over past 24 hours
- Mom reports “he sounds like a seal”
- Very uncomfortable at night
- Steeple’s Sign



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<http://www.southgatesteeplejacks.com/anatomy.html>

Summary Slide

- Knowing your anatomy will assist with your physical exam
- Your History taking will assist with your differential diagnosis
- Understanding Abnormal Physical Exam Findings will lead to your diagnosis

Reference Slide

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- **Core References:**

- Bickley, Lynn S.; Szilagyi, Peter G., “Bates' Guide to Physical Examination and History Taking”, 13e, Wolters Kluwer Health. 2023. Chapter 15: The Thorax and Lungs

<https://login.nyit.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary->

[com.nyit.idm.oclc.org/content.aspx?sectionid=248758472&bookid=2964](https://login.nyit.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=248758472&bookid=2964)

- ICC Guide, Anterior/Posterior Chest
- <https://www.practicalclinicalskills.com/lung-sounds>

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>